

7th Place Solution - single/dual+TTT

Rank	7
Team	Genius Seed is all you need
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Thank you to Kaggle and the sponsors for hosting such a wonderful and engaging competition. It was a heated contest, and I am thrilled that we managed to hold our position amidst the fierce competition. This has been an invaluable learning experience for me. Now, let me walk you through our approach:

I joined the team midway through the competition and built my work on top of the excellent baseline (LB 0.67) established by my teammates. My teammates will provide a more detailed breakdown of the baseline operations later.

My solution follows a two-stage pipeline: Stage 1 involves initial training, followed by Stage 2, which consists of online pseudo-labeling and Test-Time Training (TTT).

Stage1

model :

Single Stream Version: @shanzhong8 DINOv3-Large + Single-stream: Predicts all five components during both training and inference, using a regression head with Softplus activation. The first 80% of parameters were frozen during the training process. The overall framework follows a shared feature extractor + task-specific heads paradigm, enabling both parameter efficiency and inductive bias sharing across related regression tasks. Input resolution 1024×2048 , accommodating ultra-high-resolution imagery. Single model 5-folds without TTA and TTT can reach 0.76 public and 0.65 private LB.

Dual Stream Version: DINOv3-Large + Dual-stream: During training, the model is supervised on all five components. However, at inference time, it only predicts Dry_Green_g, Dry_Dead_g, and Dry_Clover_g, while GDM_g and Dry_Total_g are derived through post-calculation (using Softplus-activated heads). The first 80% of parameters were frozen during the training process.

loss :

r^2 loss

CV Strategy:

StratifiedGroupKFold was used with 'sampling date' as groups and 'stat' as labels. Since the original three dates were insufficient for a 5-fold split, I subdivided the WA data into seven virtual dates. A multi-threaded brute-force search across 10 million seeds was conducted to identify the split with the most balanced distribution across all five folds.

Aug (Albumentations):

Resize(single-stream: 1024×2048 ; dual-stream: 1024^2), HorizontalFlip, VerticalFlip, RandomBrightnessContrast, and ColorJitter.

Training Configuration:

1. Optimizer: AdamW with lr: $2e-4$ and wd: $1e-2$.

2. For each fold, I maintained the top 3 best-loss checkpoints and top 3 best-score (CV) checkpoints. These 6 models were then combined using SWA (Simple Weight Averaging) to enhance generalization.
3. For dual-stream models, HFlip or VFlip was applied to the integrated image before splitting it into its respective streams.

Inference Configuration:

1. A simple average of the single-stream and dual-stream model results.
 2. Only horizontal flip TTA was utilized.
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Stage2(TTT)

Due to the lack of model heterogeneity beyond DINOv3 for ensembling and the significant inference time headroom remaining, I decided to implement Test-Time Training (TTT).

First, I performed an ensemble inference of single-stream and dual-stream models (with or without TTA). To optimize efficiency, I implemented dual-GPU parallel inference, where one GPU processed the 5-fold single-stream models while the other handled the 5-fold dual-stream models. This significantly accelerated the inference process.

The initial predictions then underwent post-processing—for example, multiplying clover by 0.8 and zeroing out any component values below 0.2. These processed results were used as pseudo-labels to train the best single-stream fold (based on LB/CV) using the combined full training and test sets. This TTT stage lasted for 12 epochs, with SWA (Simple Weight Averaging) applied during the final 4 epochs to prevent overfitting and improve generalization. All other training configurations remained consistent with Stage 1.

Additionally, I attempted parallel TTT for both single-stream and dual-stream models on separate GPUs, but this did not yield any noticeable performance gains. The final result was a weighted blend of the Stage 1 and TTT predictions: $0.2 * \text{Stage 1} + 0.8 * \text{Stage 2}$.

What didn't work:

1. Model architectures other than DINOv3, such as SigLIP-2.
2. Mixup, Cutmix, and their various modified variants.
3. Auxiliary tasks and auxiliary losses.
4. DINO embeddings combined with ML models.
5. External datasets. And many other ideas that didn't provide any meaningful gain.

infer code :

[lb 0.77 | 0.65](#)

It was incredibly difficult to find ideas that provided a stable boost in this competition. Finally, thanks to everyone—it was a pleasure competing with you all!

Supplemental Q&A

BokingChen · 2026-01-29T02:11:25.910000

Thank you for sharing this so quickly ! Without Test-Time Training, what is the score for this single mode?

Boredom · 2026-01-29T02:17:30.890000

Interestingly, we had a single-stream 5-fold model that reached 0.65 on the LB, though it was still slightly lower than the scores achieved with TTT. Aside from that, all other non-TTT models (including the ensemble) consistently hovered around 0.64.

water joe · 2026-01-29T02:56:01.747000

Would you like to know what the triggering probability was for your Mixup experiment and what the value of λ was?

Boredom · 2026-01-29T03:04:48.540000

Are you saying that using mixup successfully boosted both your CV and LB? ☒

water joe · 2026-01-29T03:16:48.710000

I used it, but you said it wasn't effective. I'm not sure now because the score can only be seen in two digits, so I can't tell whether there was an actual improvement or not. At that time, I found through the experiment that the trigger probability has a significant impact on R^2 , and it should not exceed $p=0.5$. So I wanted to ask about your parameters. 😊

PursueML · 2026-01-29T03:28:47.270000

Great work ☒ In your solution, I learned about dual-GPU parallel inference and the Test-Time Training (TTT) approach. When using pseudo labels for TTT, my understanding is that you first run inference on the hidden test data, then perform post-processing, and afterward train the best-performing single-stream fold. The entire training lasts 12 epochs, with SWA applied during the last 4 epochs. Is my understanding correct? Additionally, I'm confused: were the rules for generating pseudo labels derived through multiple probes on the LB (leaderboard)? I'm doubtful about the effectiveness of this practice.

Boredom · 2026-01-29T03:38:13.617000

Thanks 😊. For the pseudo-labeling strategy, he just used a simple average of the single-stream and dual-stream models that had the best combined CV/LB performance. (He also used some post-processing like $\text{clover} * 0.8$; honestly, I'm with you—I don't quite buy it either, it feels more like a gamble haha.) > Finally, regarding the logic of training the best-performing fold on both the full training set and the pseudo-labeled test set: the goal was to let the model capture the nuances between the Ground Truth and the noisy pseudo-labels before it overfits. It's almost like an adversarial approach, in a way.

Boredom · 2026-01-29T03:45:05.230000

I used 0.5 for mixup probability and experimented with alpha values of 0.3 and 0.4, but with no luck. Might just be me, though. Maybe others have made it work.